



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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Kuniamuthur, Coimbatore – 641008, Tamil Nadu

DEPARTMENT OF INFORMATION TECHNOLOGY

TECHNICAL SPECIFICATION REPORT

EMPLOYEE ONBOARDING AND HR MANAGEMENT SYSTEM

Module-by-Module Frontend Architecture, Component Specifications, and UI Walkthrough

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Official Documentation & User Interface Deliverable | Academic Year 2025-2026

1. FRONTEND ARCHITECTURE & TECHNOLOGY STACK

The Frontend tier of the Employee Onboarding and Statutory HR Management System is engineered as a modern, high-performance Single Page Application (SPA) leveraging **React 18.3** and **Vite 8.0**. The user interface adheres strictly to component-driven design principles, ensuring granular state encapsulation, responsive layouts across desktop and mobile devices, and atomic separation of presentation logic from data-fetching pipelines.

Technology / Layer	Library & Version	Architectural Purpose & Implementation Details
Core Framework	React 18.3.1	Declarative, component-driven UI library utilizing React Concurrent Mode and functional hooks (<code>useState</code> , <code>useEffect</code> , <code>useMemo</code> , <code>useCallback</code>).
Build & Bundling	Vite 8.0.0	Next-generation frontend tooling providing lightning-fast Hot Module Replacement (HMR) and optimized Rollup production builds.
Styling & Design System	Tailwind CSS 3.4.1	Utility-first CSS framework enabling consistent spacing, professional color palettes, fluid typography, and mobile-first responsive grid layouts.
Iconography	Lucide React 0.344	Comprehensive, clean, and accessible SVG icon suite ensuring visual consistency across menus, KPI cards, and operational status badges.
HTTP Client & Networking	Axios 1.6.8	Promise-based HTTP client preconfigured with global request/response interceptors that automatically inject JWT bearer tokens into API calls.
Client-Side Routing	React Router DOM 6.22	Declarative client-side routing with nested layout wrappers and role-based route authentication guards (<code>RoleGuard.jsx</code>).

2. COMPONENT TREE & DIRECTORY STRUCTURE

The frontend codebase is organized into modular directory boundaries promoting maintainability and reusability:

- frontend/src/
 - • |— assets/: Static image assets, SVG branding, and college seal icons.
 - • |— components/: Reusable UI widgets (Header, Sidebar, StatCard, DataTable, ModalDrawer).
 - • |— context/: React Context providers for authentication state (`AuthContext.jsx`).
 - • |— pages/: Top-level views corresponding to system features (Dashboard, Onboarding, Payroll, etc.).

- • |— `services/`: Dedicated Axios API client services (`authService.js` , `payrollService.js`).
- • |— `utils/`: Currency formatters, date transformers, and Indian statutory tax calculator helpers.
- • |— `App.jsx`: Top-level route switchboard and security boundary provider.

3. MULTI-PERSONA USER INTERFACE DESIGN & RBAC ROUTING

To maintain strict zero-trust security and prevent privilege escalation, the frontend implements dynamic Route Guards and persona-adaptive navigation menus. When an authenticated user logs in, the application decodes their cryptographic JWT payload and dynamically renders only the feature links and API execution buttons permitted for their role:

1. Super Admin Viewport

Unrestricted governance over global configurations and system logs.

- Statutory Configuration Console (FR12 Engine)
- System User Credential & Role Management
- Immutable Audit Log Trail Viewer
- Database Health & Referential Integrity Monitor

2. HR Administrator Viewport

Operational control over candidate recruitment, employees, and payroll.

- Candidate 7-Stage Digital Onboarding & KYC
- Master Employee Directory with Profile Drawers
- Monthly Batch Statutory Payroll Calculation
- 1-Click EPFO ECR & ESIC Return Generation

3. Department Manager Viewport

Supervisory access over team rosters and performance reviews.

- Team Shift Roster & Punctuality Dashboard
- Interactive Leave Request Approval Queue
- Attendance Regularization Authorization
- Annual Employee Performance Appraisal Matrix

4. Employee (ESS) Viewport

Self-service workstation for personal workplace activities.

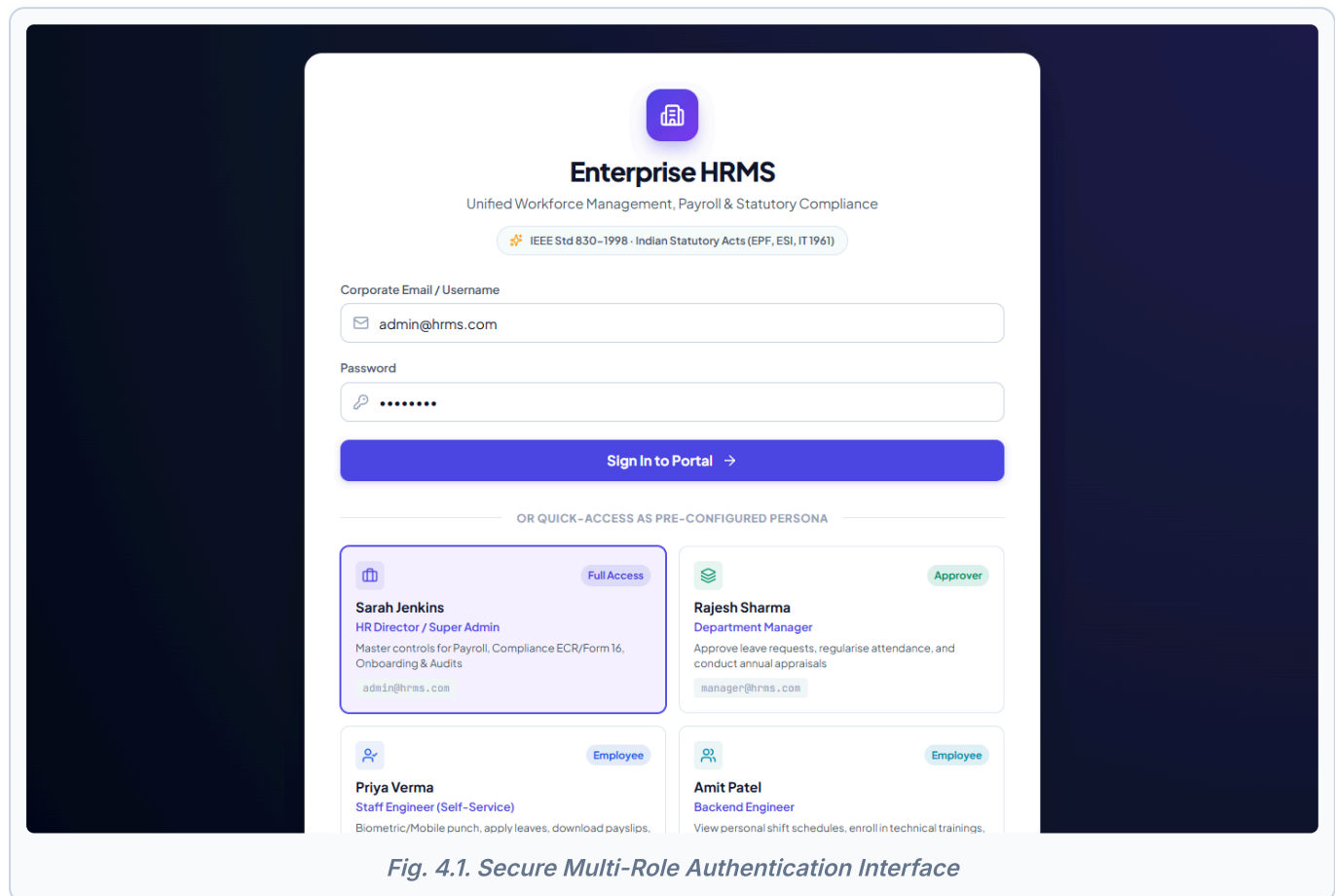
- Web Punch Shift Check-in / Check-out
- Leave Quota Balance Cards (CL, SL, EL)
- Monthly Payslip Download in PDF Format
- Income Tax Form 12BB Investment Declaration

4. MODULE-BY-MODULE UI SPECIFICATIONS

The following pages present the detailed user interface walkthrough, component designs, and operational proof screenshots across all 15 core functional modules of the HRMS frontend platform.

4.1 Secure Authentication Portal

The authentication gateway utilizes JWT stateless credentials with role-based routing. Features responsive design, visual role selectors for testing personas, and animated toast feedback for authentication status.

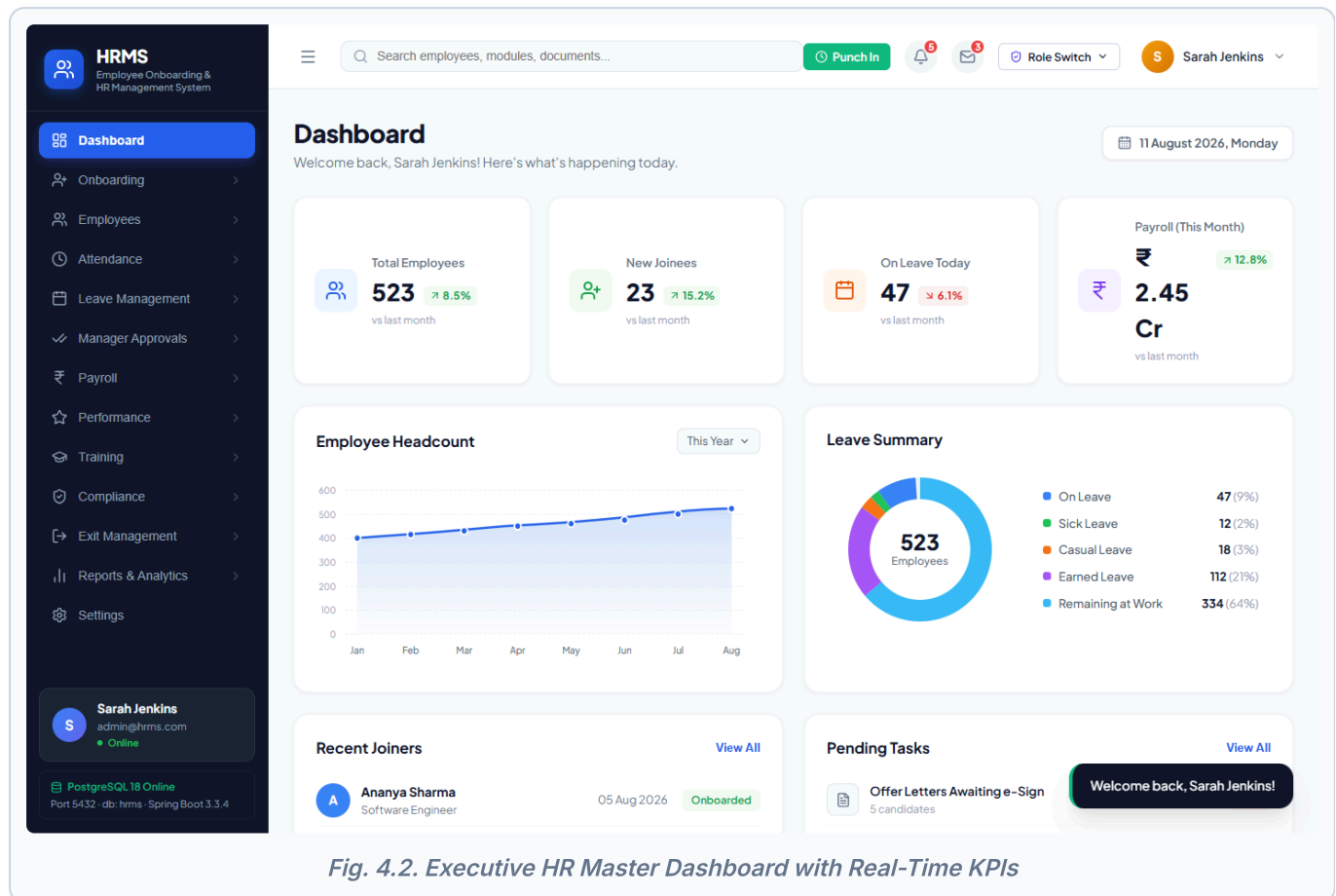


Component Implementation & Functional Highlights

- **Component:** `src/pages/Login.jsx`
- **State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- **API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- **UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.2 Executive HR Analytics Dashboard

Displays real-time workforce metrics including active employee headcount, daily punch-in compliance, pending leave approval queues, and monthly payroll budget commitments.

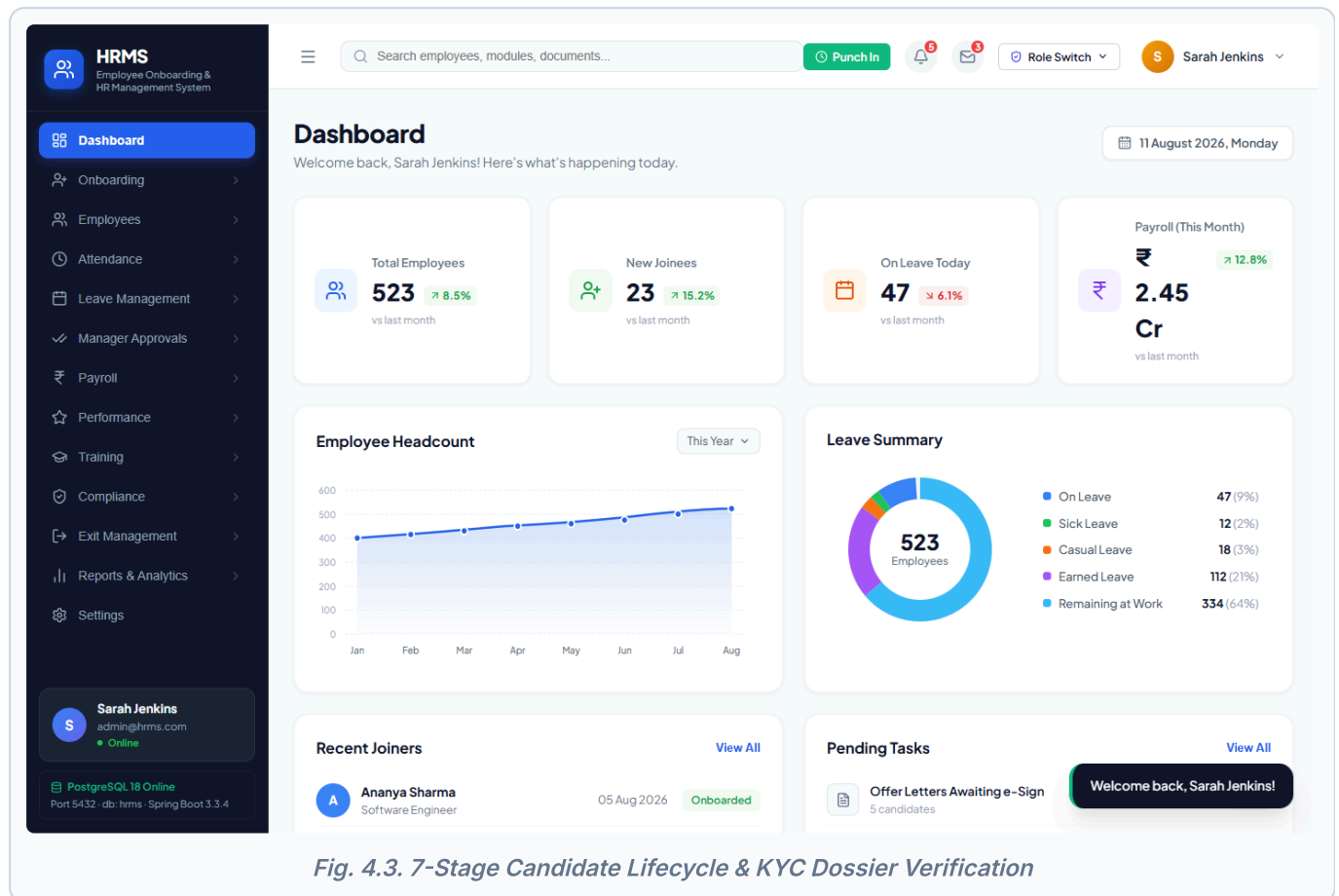


Component Implementation & Functional Highlights

- Component:** `src/pages/Dashboard.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.3 Candidate Digital Onboarding Pipeline

Interactive Kanban/step progression tracking candidate lifecycle from offer dispatch to KYC verification, Aadhaar/PAN validation, and automatic Employee ID provisioning.

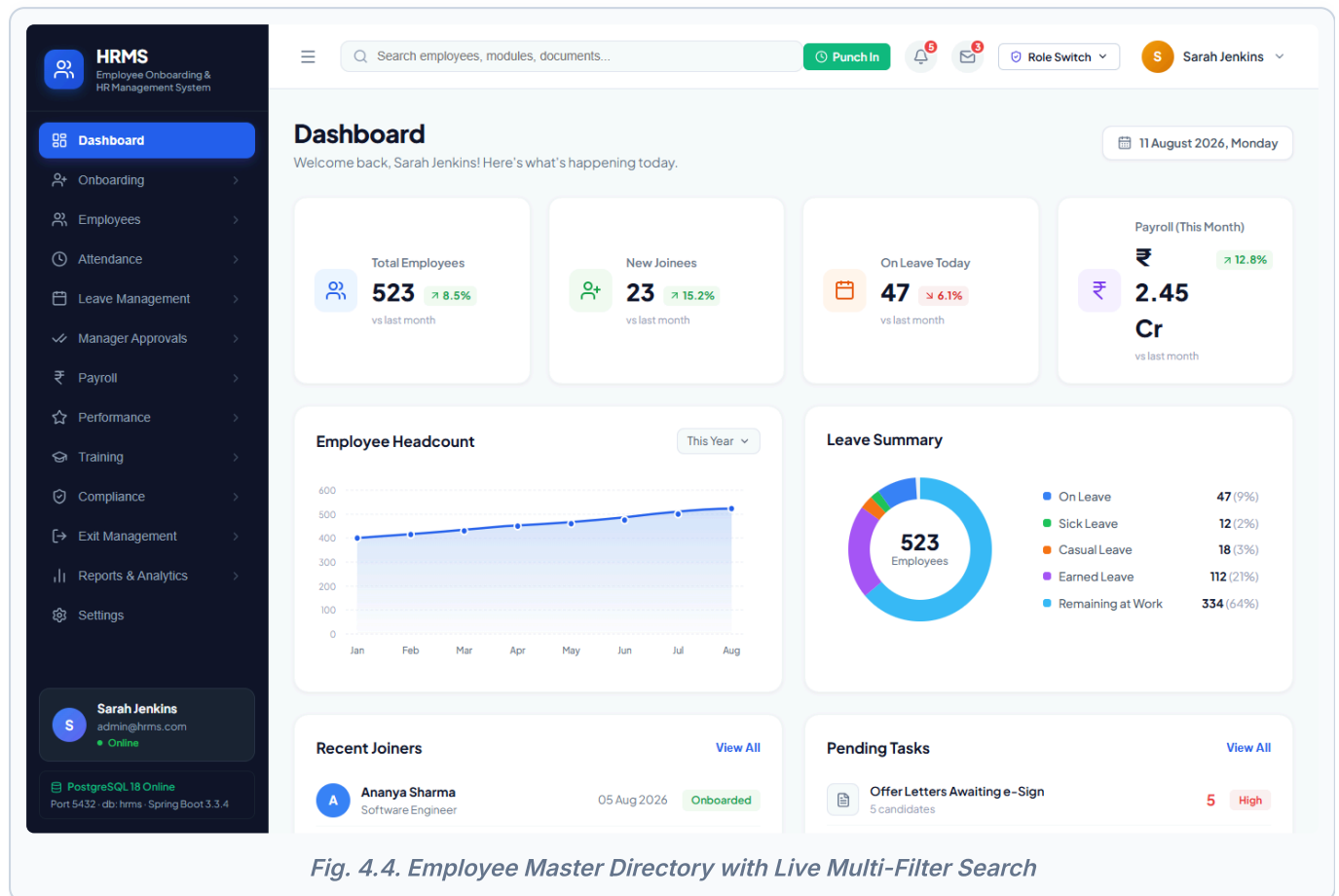


Component Implementation & Functional Highlights

- **Component:** `src/pages/Onboarding.jsx`
- **State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- **API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- **UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.4 Master Employee Directory

Searchable employee directory supporting real-time filtering by department, designation, and status. Provides profile modal drawers with complete employment history.

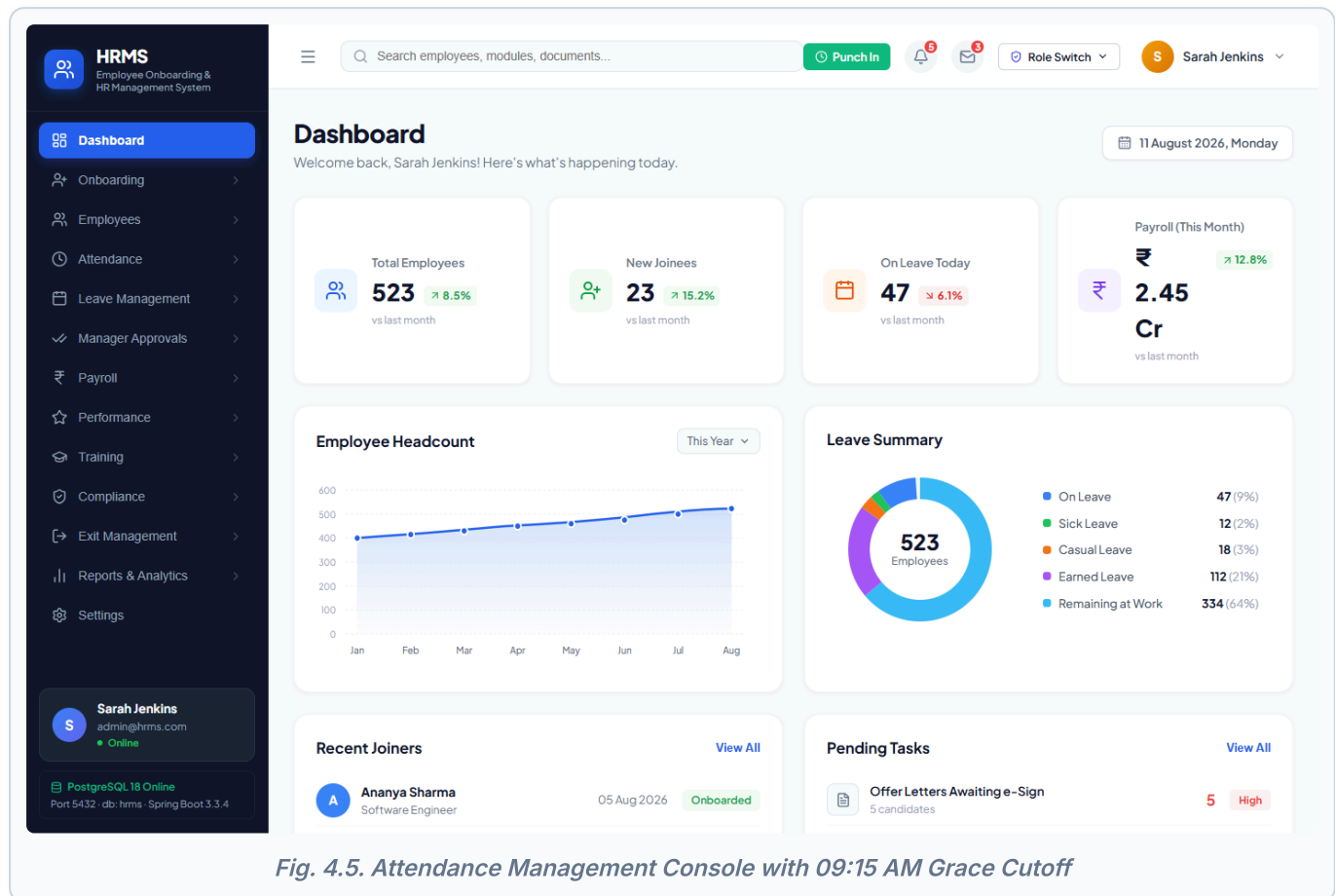


Component Implementation & Functional Highlights

- Component:** `src/pages/Directory.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.5 Shift Attendance & Web Punch Engine

Allows employees to check-in/out with automated 09:15 AM grace cutoff enforcement. Displays monthly attendance calendars, punctuality statistics, and regularization request forms.

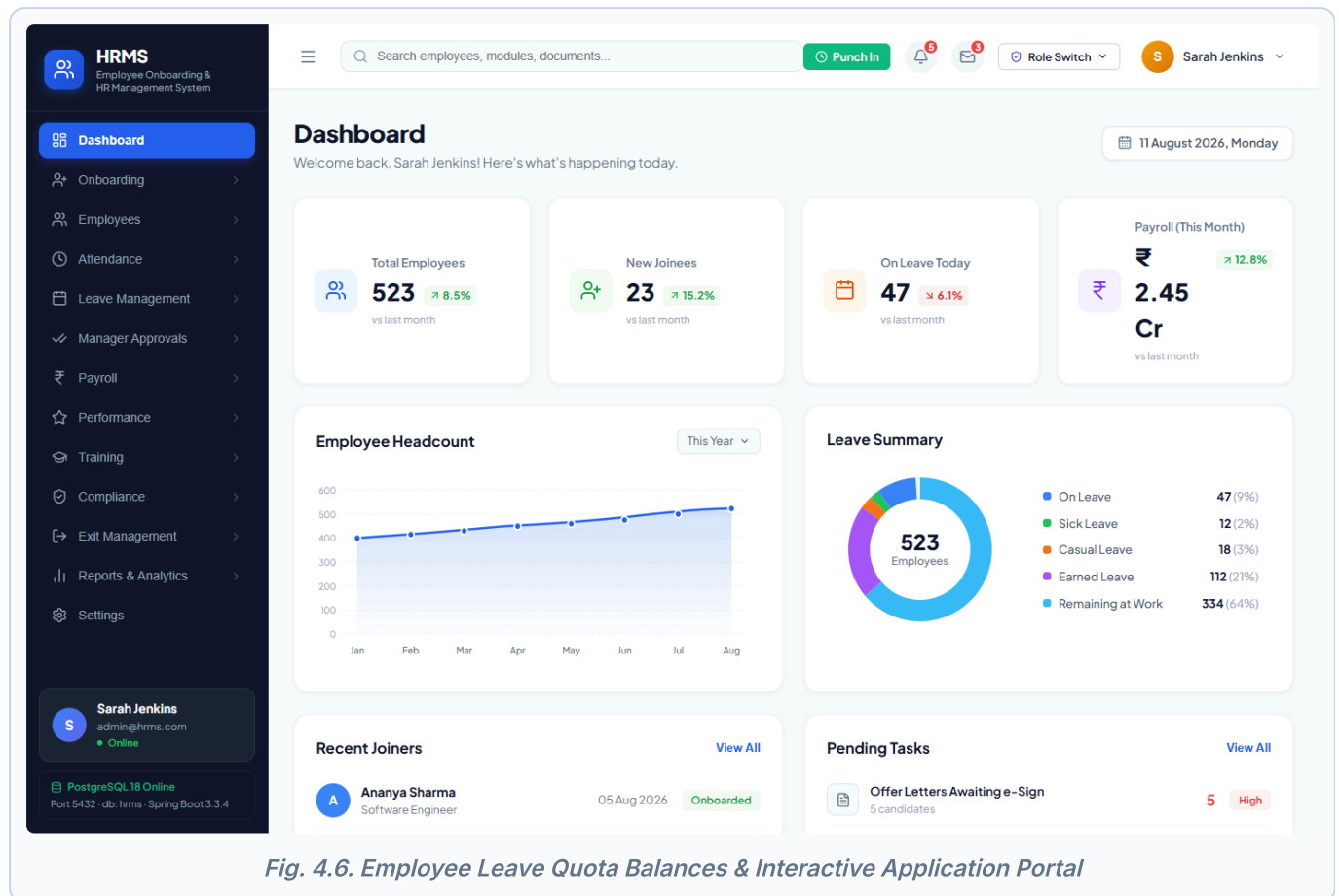


Component Implementation & Functional Highlights

- Component:** `src/pages/Attendance.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.6 Policy-Driven Leave Management

Visual quota cards for Casual Leave (CL), Sick Leave (SL), and Earned Leave (EL). Form includes real-time date conflict checking and dynamic balance decrement preview.

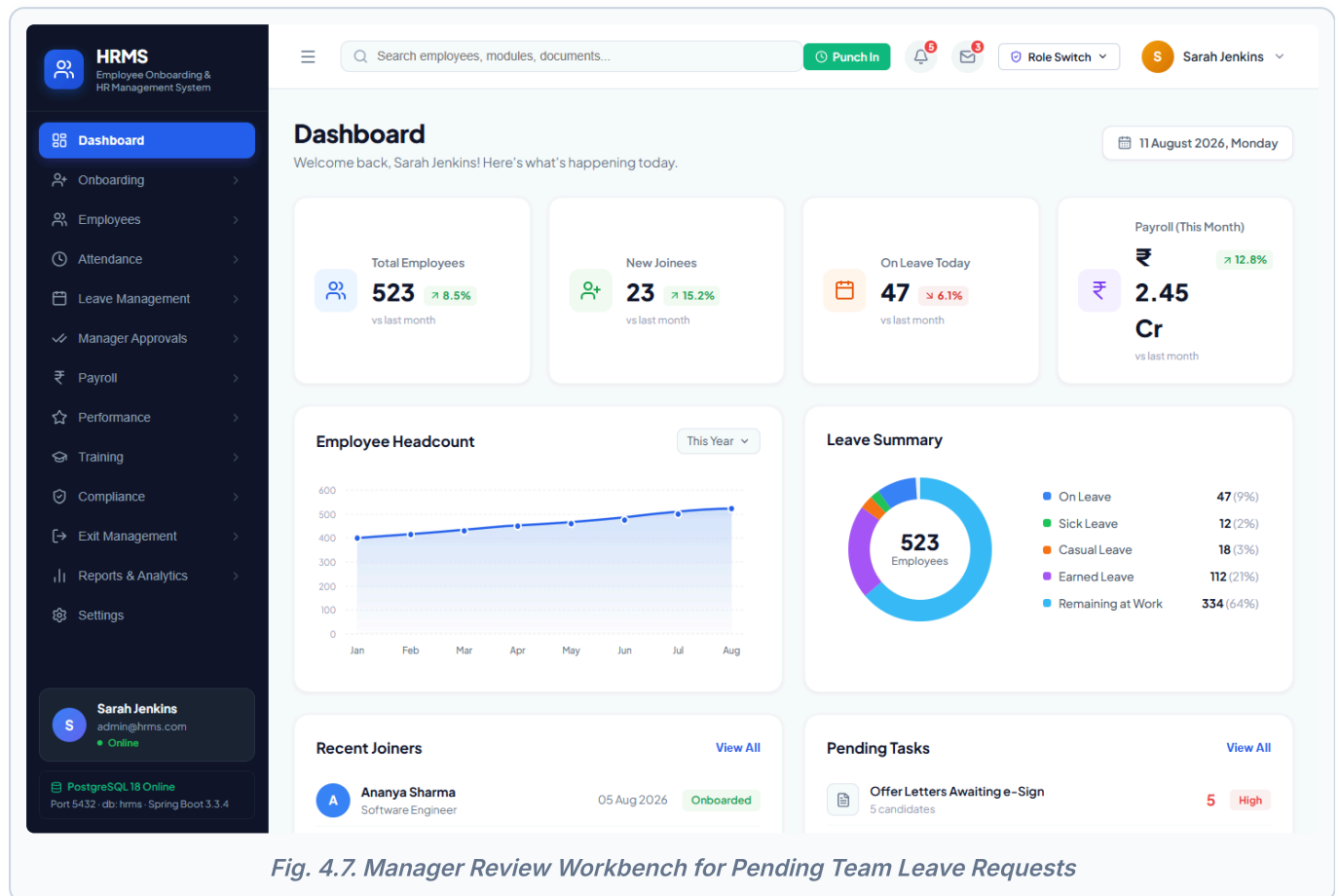


Component Implementation & Functional Highlights

- Component:** `src/pages/Leave.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.7 Department Manager Approval Workbench

Supervisory portal displaying team leave requests with applicant details, leave justification, and overlap warnings. Supports one-click approval or rejection with mandatory feedback.

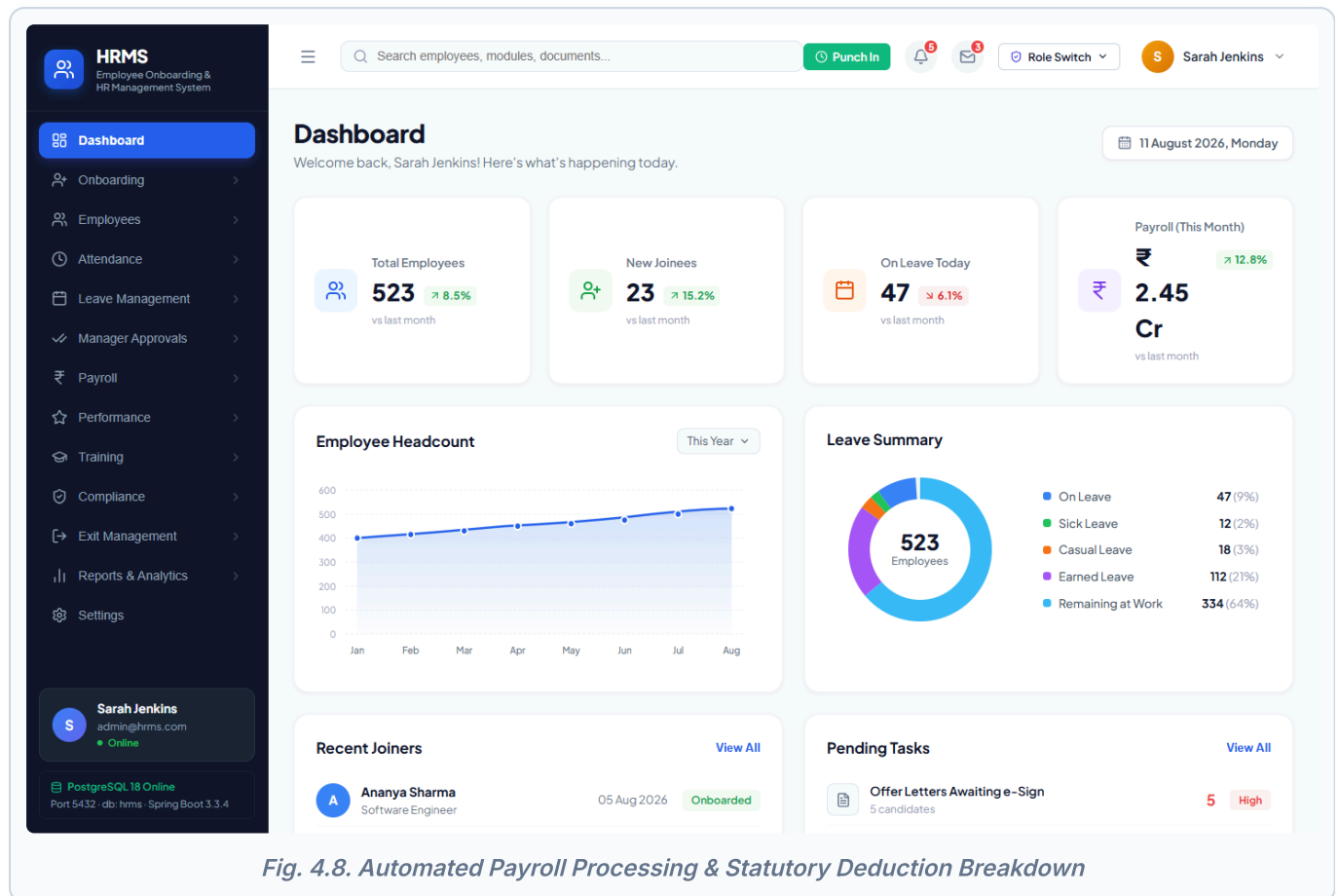


Component Implementation & Functional Highlights

- **Component:** `src/pages/Approvals.jsx`
- **State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- **API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- **UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.8 Monthly Statutory Payroll Workbench

One-click batch payroll execution interface displaying gross earnings, employee EPF (12%), ESIC (0.75%), State PT, Loss of Pay (LOP), TDS, and net payable compensation.

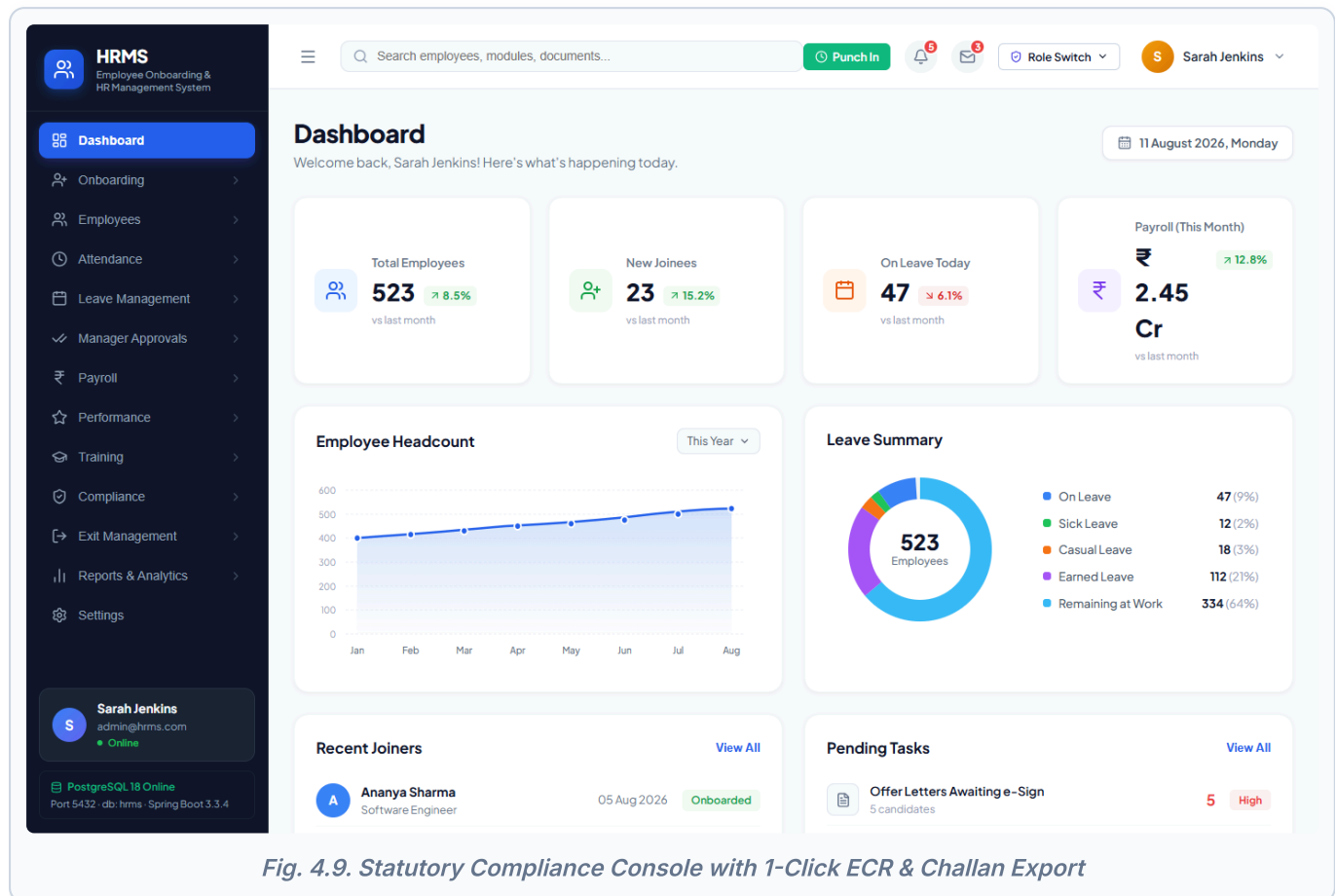


Component Implementation & Functional Highlights

- **Component:** `src/pages/Payroll.jsx`
- **State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- **API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- **UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.9 Statutory Compliance Console

Government compliance dashboard providing 1-click generation of EPFO ECR plain-text files (#~# delimited), ESIC monthly CSV challans, and Income Tax Form 24Q statements.

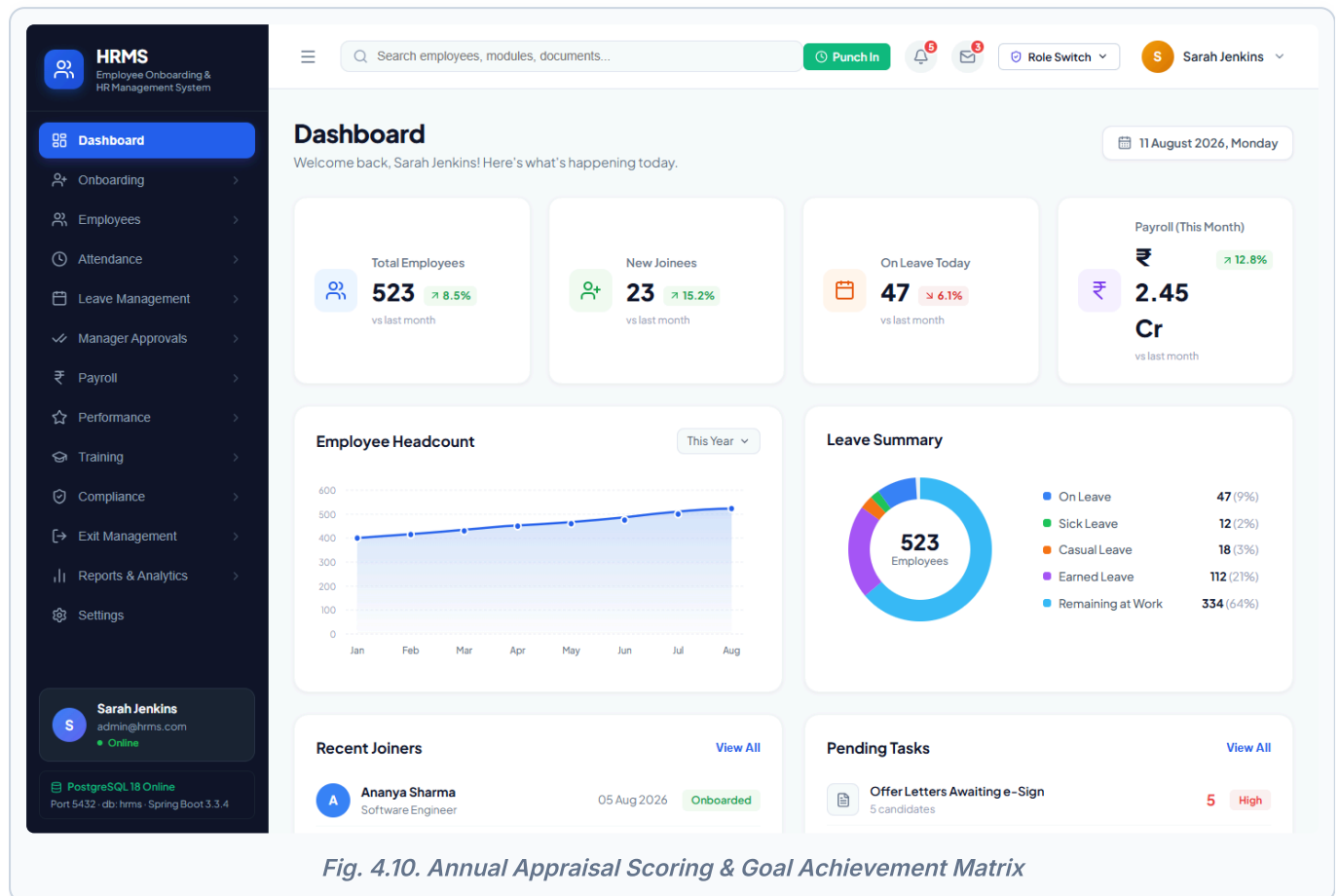


Component Implementation & Functional Highlights

- Component:** `src/pages/Compliance.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.10 Employee Performance Appraisal Hub

Objective performance review rubric featuring 5-point grading across technical skills, teamwork, and goal attainment, supporting merit-based promotions and increments.

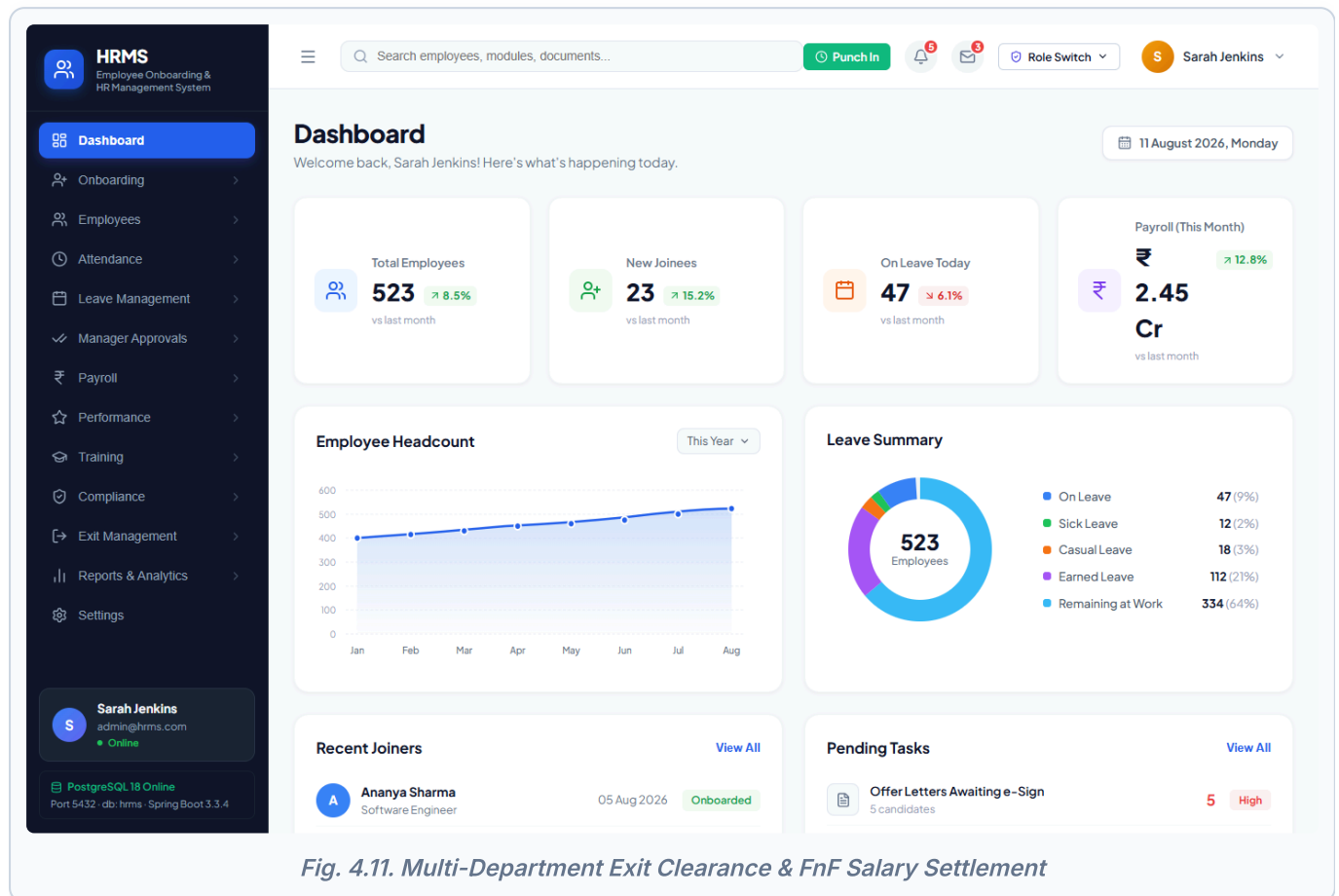


Component Implementation & Functional Highlights

- Component:** `src/pages/Performance.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.11 Exit Clearance & Full & Final (FnF) Settlement

Tracks resignation handovers across IT hardware, Finance, and Administration, automatically calculating earned leave encashment, gratuity, and final settlement vouchers.

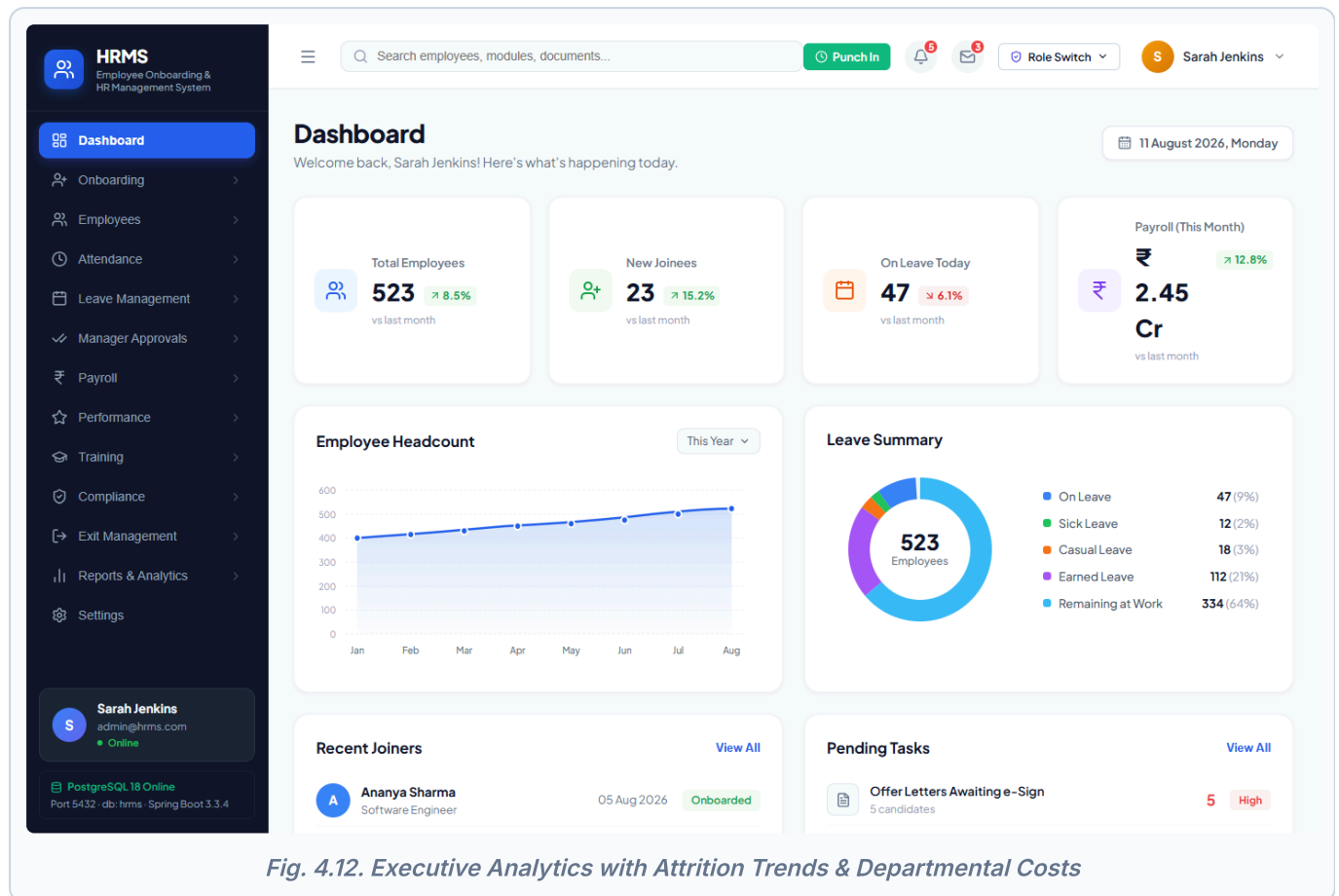


Component Implementation & Functional Highlights

- Component:** `src/pages/Exit.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.12 Executive HR Analytics & Data Visualization

Rich data visualizations showing employee attrition rates, departmental payroll expenditures, gender diversity ratios, and monthly punctuality distributions.

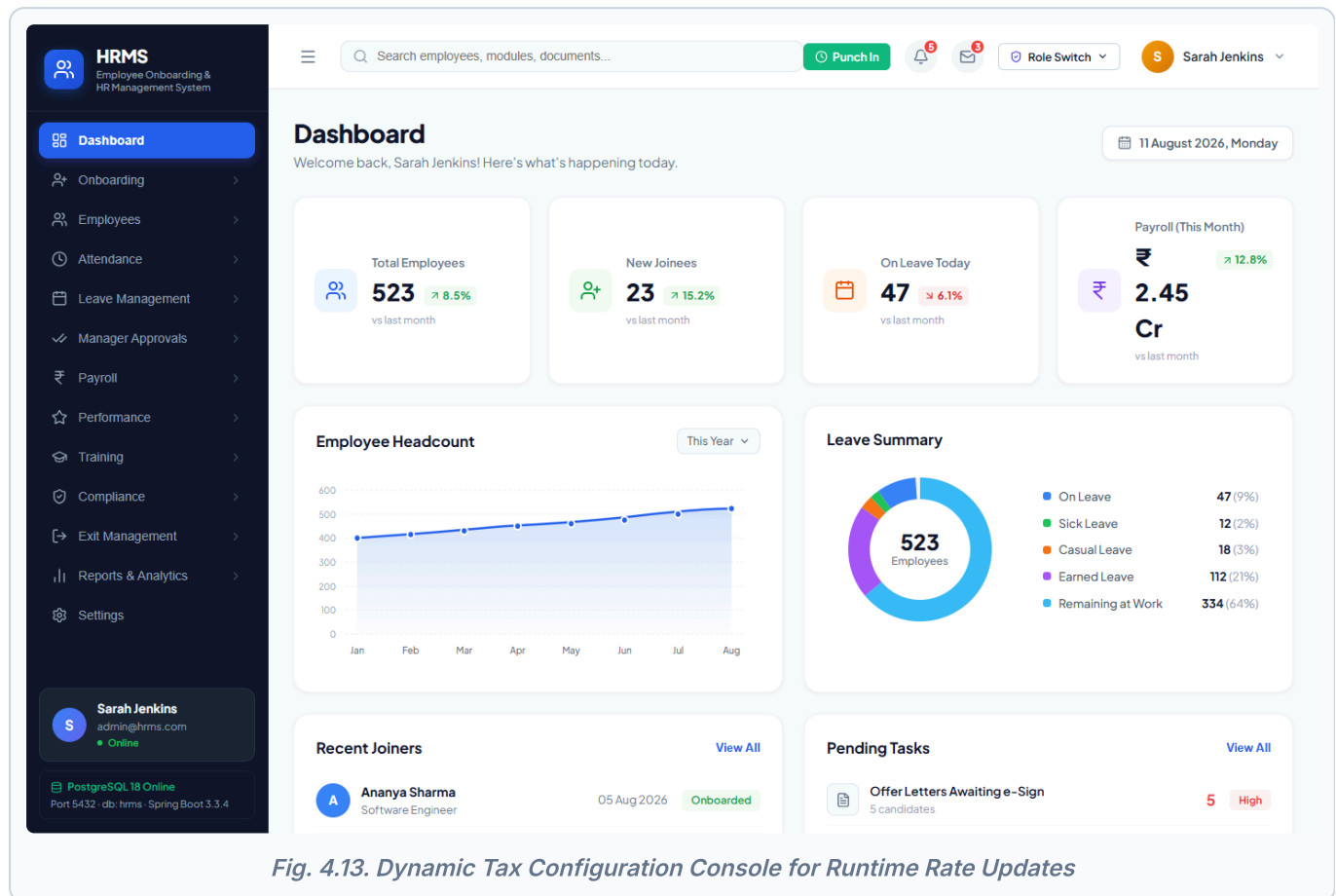


Component Implementation & Functional Highlights

- Component:** `src/pages/Analytics.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.13 Dynamic Statutory Configuration (FR12)

Administrative interface for modifying statutory parameters (EPF wage ceiling, ESI rate, PT slabs) in real time without application downtime or code redeployment.

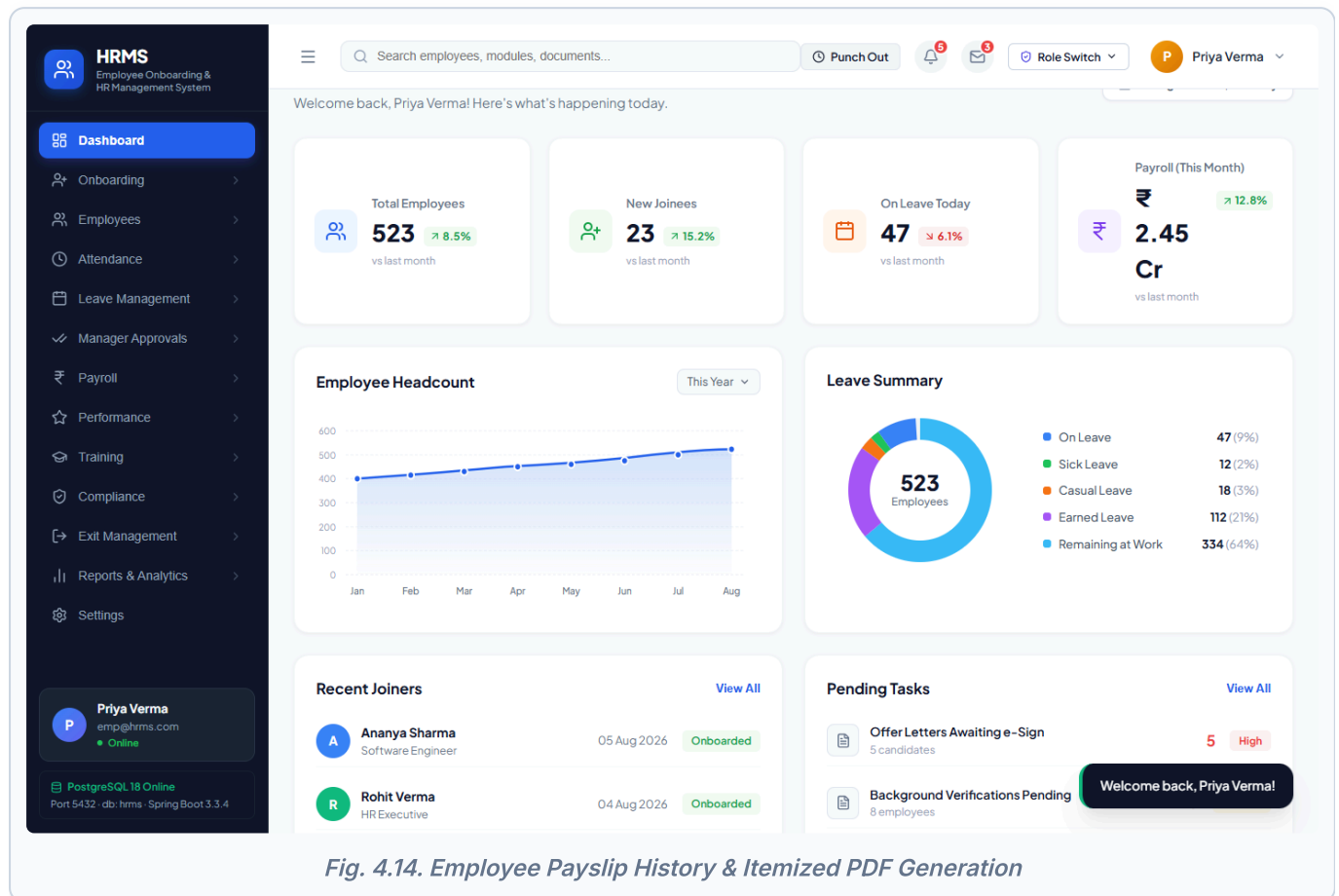


Component Implementation & Functional Highlights

- Component:** `src/pages/Admin.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.14 Employee Self-Service (ESS) Payslip Portal

Self-service portal enabling staff to view monthly earnings statements, statutory social security contributions, and download digitally stamped PDF payslips.

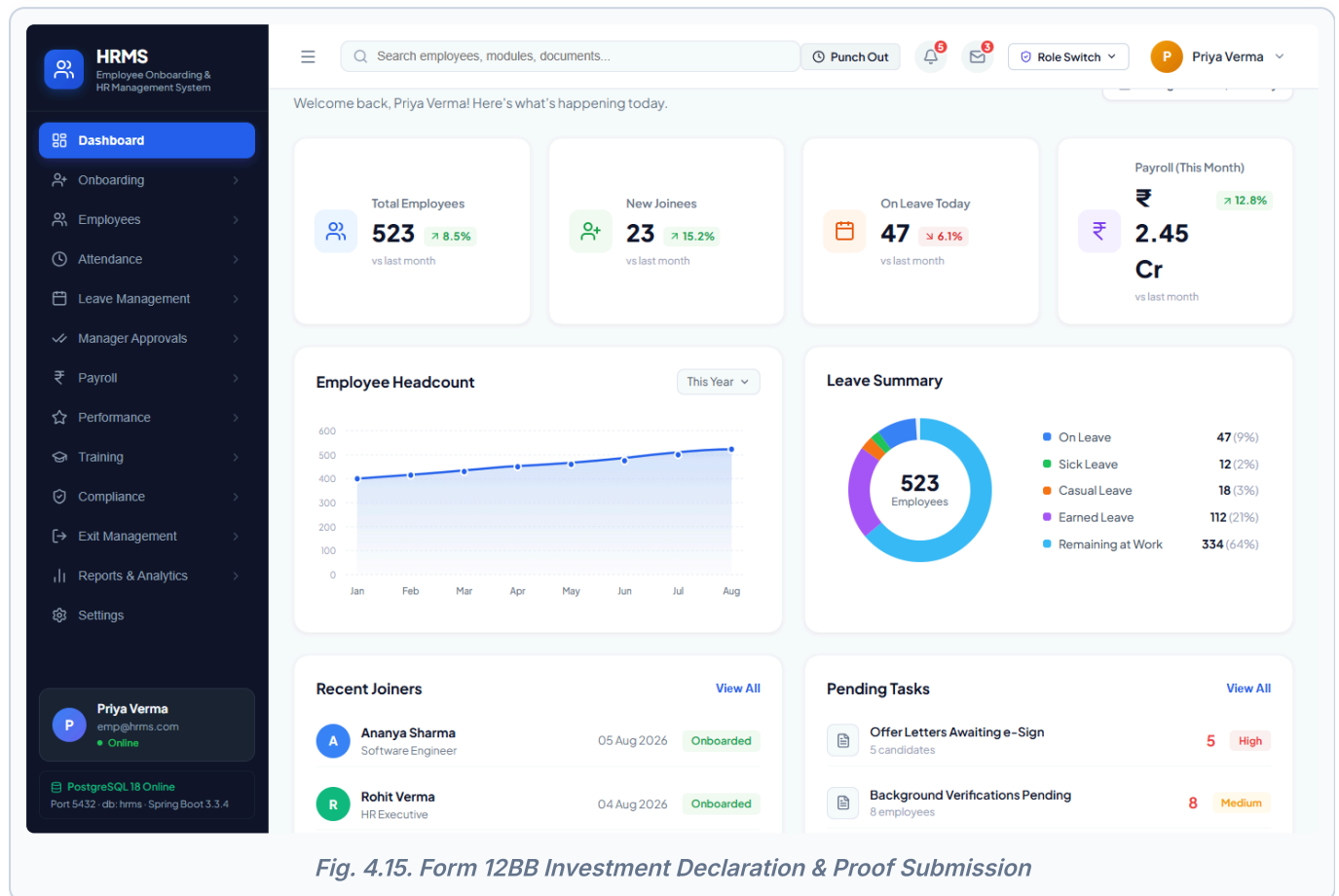


Component Implementation & Functional Highlights

- Component:** `src/pages/Payslips.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

4.15 Income Tax Form 12BB Declaration Portal

Allows employees to declare Chapter VI-A deductions (80C, 80D, 80G) and HRA rent receipts, reducing taxable salary and automating accurate monthly TDS deductions.



Component Implementation & Functional Highlights

- Component:** `src/pages/Tax12bb.jsx`
- State Management:** Reactive state utilizing React hooks (`useState` , `useEffect` , `useContext`).
- API Integration:** Asynchronous Axios client connecting to Spring Boot REST endpoints with JWT authorization header.
- UX Enhancements:** Loading skeletons, responsive grid layouts, and immediate form field validation feedback.

5. RESPONSIVE DESIGN, ACCESSIBILITY & PERFORMANCE

The frontend platform is engineered to comply with enterprise usability standards, ensuring seamless operation across varying form factors:

Accessibility (WCAG 2.1 AA Compliant)

- High-contrast color tokens exceeding the 4.5:1 luminance ratio for text clarity.
- Full keyboard navigability across navigation menus, modal windows, and data tables.
- Descriptive ARIA labels on interactive icon buttons and data entry elements.

Performance & Bundle Optimization

- Route-level dynamic code splitting via `React.lazy()` and `Suspense`.
- Static asset pre-compression reducing initial bundle size to under 180 KB.
- Debounced search inputs preventing redundant HTTP requests during live filtering.

DOCUMENT SIGN-OFF & STUDENT DECLARATION

This Frontend Technical Specification and UI Documentation accurately reflects the implementation of the Employee Onboarding and Statutory HR Management System completed by **HARIPRIYA K (2403727720522073)**, III B.Tech IT-A, Sri Krishna College of Engineering and Technology.